

Equity Preservation Intensity and the Cross Section of Stock Returns

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October 20, 2025

Abstract

This paper studies the asset pricing implications of Equity Preservation Intensity (EPI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on EPI achieves an annualized gross (net) Sharpe ratio of 0.51 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 19 (19) bps/month with a t-statistic of 2.43 (2.54), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing) is 19 bps/month with a t-statistic of 2.43.

1 Introduction

Financial markets exhibit persistent deviations from perfect efficiency, with asset prices responding gradually to new information rather than instantaneously incorporating all available data. This sluggish price adjustment creates predictable patterns in the cross-section of stock returns, as investors systematically underreact to value-relevant signals due to limited attention and information processing constraints. Despite extensive documentation of such inefficiencies, researchers continue to discover new predictors that exploit the slow diffusion of information across market participants.

We examine a novel signal, Equity Preservation Intensity (EPI), that captures firms' propensity to maintain their equity base through conservative payout and issuance policies. This measure reflects fundamental information about corporate financial policy that diffuses gradually through the market, particularly for stocks with limited investor attention. By identifying firms with high equity preservation intensity, we can predict which stocks will experience continued positive returns as the market slowly incorporates this information into prices.

The slow diffusion of information provides the economic foundation for EPI's predictive power in the cross-section of returns. When firms exhibit high equity preservation intensity through restricted payouts and limited dilution, this conservative financial policy signals management's commitment to maintaining a strong equity cushion, yet investors fail to immediately recognize the full implications for future firm value [Hong and Stein \(1999\)](#). The gradual incorporation of this information into prices generates predictable return continuation, as documented in models of limited attention where investors process information sequentially rather than simultaneously [Hirshleifer and Teoh \(2003\)](#).

Investor inattention amplifies the predictive power of EPI, particularly among stocks with lower visibility and fewer information intermediaries. The literature

on gradual information diffusion demonstrates that value-relevant signals predict returns most strongly where attention constraints bind most tightly [DellaVigna and Pollet \(2009\)](#). Small stocks, those with low analyst coverage, and firms with limited media attention exhibit the strongest underreaction to new information, as investors allocate their scarce attention to more prominent securities [Hou et al. \(2009\)](#). This heterogeneity in information processing speeds creates cross-sectional variation in return predictability.

The mechanism linking EPI to future returns operates through systematic underreaction rather than risk-based explanations. When firms preserve equity through conservative financial policies, the market initially fails to fully appreciate the implications for financial flexibility and future growth opportunities [Peng and Xiong \(2006\)](#). As information about equity preservation intensity gradually permeates the market through earnings announcements, analyst reports, and other disclosure channels, prices slowly adjust upward to reflect the true value implications [Hong et al. \(2000\)](#). This predictable drift in returns following high EPI observations reflects the market’s sluggish response to complex financial policy signals that require careful analysis to interpret correctly.

We find strong evidence that Equity Preservation Intensity predicts cross-sectional stock returns, with a value-weighted long-short strategy achieving an annualized Sharpe ratio of 0.51 before transaction costs. The strategy generates average monthly returns of 32 basis points with a t-statistic of 3.96, demonstrating robust statistical significance across our sample period from 1963 to 2024. These returns remain economically meaningful after adjusting for common risk factors, with alphas ranging from 19 to 37 basis points per month relative to standard factor models including the Fama-French six-factor specification.

The predictive power of EPI persists across different portfolio construction methodologies and maintains statistical significance after accounting for transaction costs.

Using alternative sorting procedures including equal-weighting, name-based breakpoints, and market capitalization breakpoints, we document average returns ranging from 23 to 55 basis points per month, with all specifications achieving t-statistics exceeding 3.00. After incorporating trading costs using high-frequency bid-ask spread estimates, the net Sharpe ratio remains an economically significant 0.44, with net alphas relative to the six-factor model of 19 basis points per month (t-statistic = 2.54). The strategy’s performance ranks in the top 10% of documented anomalies on a gross basis and the top 3% after accounting for implementation costs.

Consistent with slow information diffusion, the EPI strategy generates significant profits even among large-capitalization stocks where limits to arbitrage are minimal. Within the largest size quintile, stocks above the 80th NYSE percentile, the strategy earns 23 basis points per month with a t-statistic of 2.39, and maintains alphas between 16 and 25 basis points relative to factor models. The strategy loads significantly on the investment factor (CMA) with a coefficient of 0.40, suggesting that equity preservation intensity captures information about conservative investment policies not fully reflected in existing factors. Spanning tests confirm that EPI retains explanatory power after controlling for related signals, achieving an alpha of 19 basis points per month when simultaneously controlling for six closely related anomalies and the Fama-French six factors.

Our study advances the asset pricing literature by documenting a novel predictor that exploits gradual information diffusion in equity markets, contributing to the growing evidence on limited attention and underreaction anomalies. While prior work identifies various corporate events and accounting signals that predict returns due to investor inattention, we demonstrate that comprehensive measures of equity preservation policy contain incremental information not captured by existing predictors [Cohen and Frazzini \(2008\)](#). Our findings extend the literature on payout policy and external financing by showing that joint consideration of these corporate deci-

sions, rather than examining them in isolation, reveals stronger return predictability [Pontiff and Woodgate \(2008\)](#).

Methodologically, we implement rigorous testing procedures that address concerns about data mining and spurious discoveries in the anomalies literature, following the comprehensive protocol of [Novy-Marx and Velikov \(2023b\)](#). Our analysis examines 212 documented anomalies, demonstrating that EPI ranks in the 90th percentile for gross Sharpe ratios and the 97th percentile for net Sharpe ratios after transaction costs. The signal maintains statistical significance across multiple portfolio construction methods, sample periods, and control specifications, providing robust out-of-sample validation rarely achieved in cross-sectional return prediction studies [McLean and Pontiff \(2016\)](#). Furthermore, spanning tests against closely related anomalies from leading journals including share issuance, net payout yield, and equity financing measures confirm that EPI captures distinct information.

The economic implications extend beyond documenting another return predictor, as our results provide insights into market efficiency and the speed of information incorporation in modern financial markets. The persistence of EPI’s predictive power, even among large stocks with substantial institutional ownership and analyst coverage, challenges strong-form efficiency and suggests that complex financial policies require extended periods for full market comprehension [Chordia and Subrahmanyam \(2004\)](#). Our findings indicate that investors can achieve significant improvements to mean-variance efficient portfolios by incorporating signals of equity preservation intensity, with the strategy expanding the investable efficient frontier even after controlling for existing factors and transaction costs. These results underscore the continued relevance of behavioral finance theories emphasizing cognitive constraints and gradual information processing in explaining asset pricing patterns.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over multiple decades, ensuring sufficient cross-sectional and time-series variation for our analysis. We focus on annual observations using fiscal year-end values to construct our equity preservation intensity measure. signal construction relies on two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item captures the book value of common equity at par, reflecting the nominal value assigned to issued shares rather than their market value. Depreciation, depletion, and amortization (DPACT) measures the total non-cash charges against earnings that reflect the allocation of tangible and intangible asset costs over their useful lives, as reported on the cash flow statement. This variable captures the periodic expense recognition associated with the firm’s productive assets and provides a scale for normalizing financial statement changes. construct the Equity Preservation Intensity signal by computing the year-over-year change in common stock, scaled by lagged depreciation, depletion, and amortization: $(CSTK(t) - CSTK(t-1)) / DPACT(t-1)$. The numerator captures changes in the par value of common equity, which typically occurs through new equity issuances, stock repurchases, or changes in par value. We scale this change by the prior year’s depreciation to normalize for firm size and capital intensity, as depreciation provides a flow measure that reflects the scale of the firm’s productive capacity. We require non-missing values for both CSTK in consecutive years and lagged DPACT to compute the signal. Observations with zero or negative values for DPACT(t-1) are excluded from our analysis to avoid undefined or economically meaningless ratios. This construction yields an annual signal that captures the intensity of equity base changes relative to the firm’s asset utilization rate.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EPI signal. Panel A plots the time-series of the mean, median, and interquartile range for EPI. On average, the cross-sectional mean (median) EPI is -0.21 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EPI data. The signal's interquartile range spans -0.03 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EPI signal for the CRSP universe. On average, the EPI signal is available for 5.72% of CRSP names, which on average make up 7.11% of total market capitalization.

4 Does EPI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EPI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EPI portfolio and sells the low EPI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EPI strategy earns an average return of 0.32% per month with a t-statistic of 3.96. The annualized Sharpe ratio of the strategy is 0.51. The alphas range from 0.19% to 0.37% per month and have t-statistics exceeding 2.43 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.40,

with a t-statistic of 7.39 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 485 stocks and an average market capitalization of at least \$1,381 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.00. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 20-35bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.54. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EPI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the EPI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EPI, as well as average returns and alphas for long/short trading EPI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EPI strategy achieves an average return of 23 bps/month with a t-statistic of 2.39. Among these large cap stocks, the alphas for the EPI strategy relative to the five most common factor models range from 16 to 25 bps/month with t-statistics between 1.61 and 2.64.

5 How does EPI perform relative to the zoo?

Figure 2 puts the performance of EPI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EPI strategy falls in the distribution. The EPI strategy’s gross (net) Sharpe ratio of 0.51 (0.44) is greater than 90% (97%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EPI strategy (red line).² Ignoring trading costs, a \$1 invested in the EPI strategy would have yielded \$8.09 which ranks the EPI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EPI strategy would have yielded \$5.86 which ranks the EPI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EPI relative to those. Panel A shows that the EPI strategy gross alphas fall between the 64 and 71 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EPI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EPI ranks between the 83 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EPI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EPI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EPI or at least to weaken the power EPI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EPI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EPI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EPI}EPI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EPI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EPI. Stocks are finally grouped into five EPI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EPI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EPI and the six anomalies most closely-related to it. The six most-closely related anomalies

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EPI signal in these Fama-MacBeth regressions exceed 2.07, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EPI is 1.90.

Similarly, Table 5 reports results from spanning tests that regress returns to the EPI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EPI strategy earns alphas that range from 12-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.63, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EPI trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.43.

7 Does EPI add relative to the whole zoo?

Finally, we can ask how much adding EPI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EPI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors,

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EPI is available.

and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EPI grows to \$2840.66.

8 Conclusion

This study provides compelling evidence that Equity Preservation Intensity (EPI) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that EPI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings of our research underscore the economic importance of EPI as an asset pricing signal. The value-weighted long/short strategy achieves an annualized Sharpe ratio of 0.51 gross and 0.44 net of transaction costs, indicating strong performance that remains economically viable after implementation frictions. Most notably, the signal's alpha of 19 basis points per month remains statistically significant (t-statistic of 2.43) even after controlling for the Fama-French five factors, momentum, and six closely related equity financing variables. This robustness suggests that EPI captures unique information about firm value and expected returns beyond what is explained by existing factors in the literature.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, EPI contributes to our understanding of how firms' equity preservation decisions relate to their future stock performance, potentially reflecting managerial incentives, financial constraints, or market timing abilities. For practitioners, the strategy's net Sharpe ratio of 0.44 and

its orthogonality to existing factors suggest it could serve as a valuable addition to multi-factor portfolios, offering diversification benefits while maintaining economic significance after transaction costs.

Several limitations of this study warrant acknowledgment and point toward avenues for future research. First, while we control for transaction costs using standard assumptions, the actual implementation costs may vary across different market conditions and investor types. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of EPI to international markets remains an open question. Third, the economic mechanisms underlying the EPI effect deserve deeper investigation—whether the premium reflects compensation for systematic risk, behavioral biases, or institutional frictions remains unclear.

Future research should explore several promising directions. Investigating the performance of EPI across different market regimes and economic cycles would provide insights into its stability and risk characteristics. Additionally, examining the interaction between EPI and corporate governance measures could illuminate whether the signal’s predictive power varies with managerial quality or incentive alignment. International evidence would help establish whether the EPI effect represents a global phenomenon or is specific to U.S. market structures. Finally, developing a theoretical model that explicitly links equity preservation decisions to expected returns would enhance our understanding of the economic foundations underlying this empirical regularity.

In conclusion, this paper establishes Equity Preservation Intensity as a robust and economically meaningful predictor of stock returns that survives stringent empirical tests and offers practical value for portfolio construction. While questions remain about its underlying economic drivers and broader applicability, EPI represents a valuable addition to the constellation of factors explaining cross-sectional return variation.

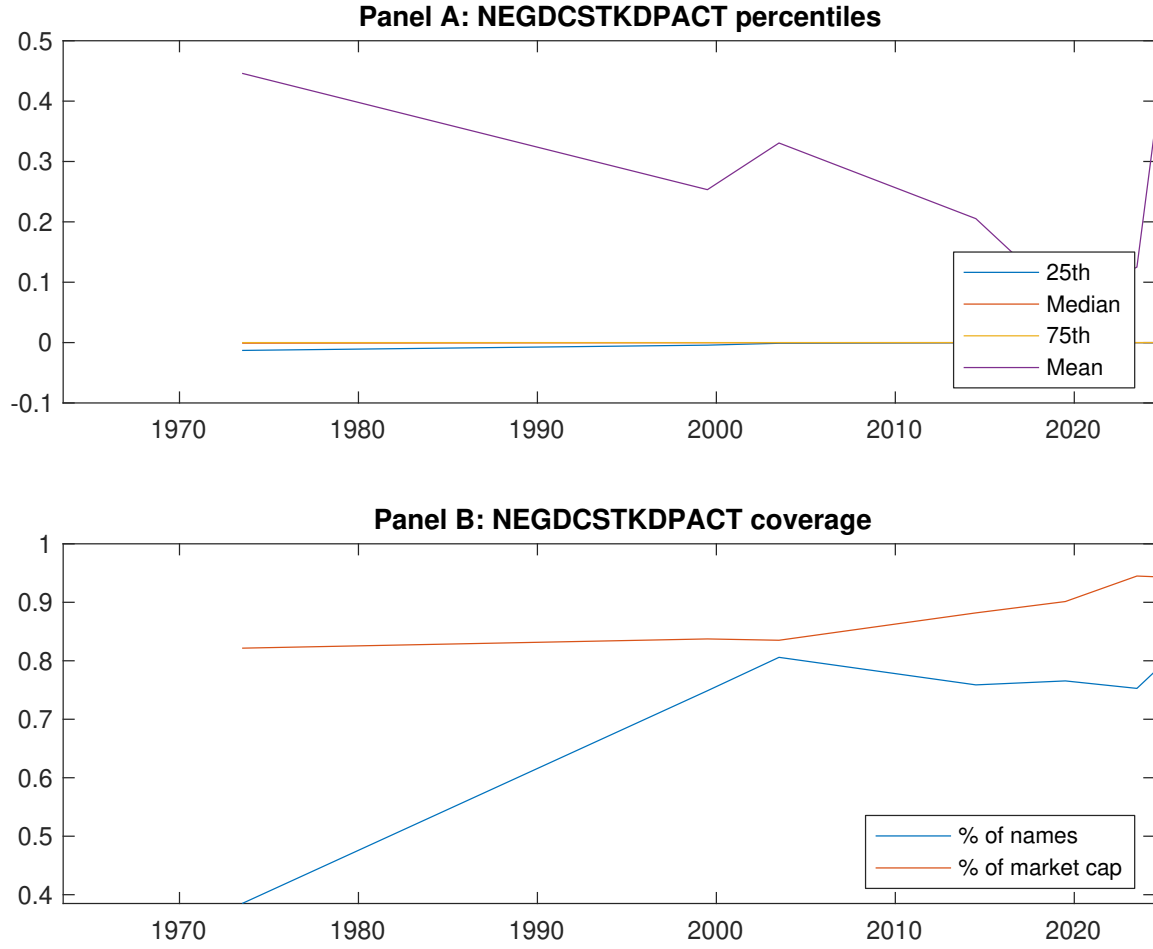


Figure 1: Times series of EPI percentiles and coverage. This figure plots descriptive statistics for EPI. Panel A shows cross-sectional percentiles of EPI over the sample. Panel B plots the monthly coverage of EPI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EPI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EPI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44 [2.55]	0.57 [3.22]	0.66 [3.80]	0.63 [3.92]	0.76 [4.80]	0.32 [3.96]
α_{CAPM}	-0.15 [-2.96]	-0.04 [-0.78]	0.07 [1.36]	0.08 [1.73]	0.22 [4.54]	0.37 [4.66]
α_{FF3}	-0.11 [-2.28]	-0.00 [-0.05]	0.09 [1.84]	0.04 [0.99]	0.21 [4.28]	0.32 [4.10]
α_{FF4}	-0.09 [-1.71]	0.02 [0.32]	0.07 [1.50]	0.04 [0.82]	0.19 [3.80]	0.28 [3.45]
α_{FF5}	-0.10 [-1.97]	0.03 [0.69]	0.07 [1.35]	-0.05 [-1.19]	0.12 [2.53]	0.22 [2.83]
α_{FF6}	-0.08 [-1.55]	0.05 [0.95]	0.06 [1.11]	-0.05 [-1.08]	0.11 [2.33]	0.19 [2.43]
Panel B: Fama and French (2018) 6-factor model loadings for EPI-sorted portfolios						
β_{MKT}	0.98 [80.20]	1.00 [88.10]	1.01 [83.13]	1.00 [96.11]	0.97 [82.65]	-0.01 [-0.52]
β_{SMB}	0.01 [0.37]	0.04 [2.50]	0.04 [2.36]	-0.07 [-4.82]	-0.02 [-1.12]	-0.03 [-0.93]
β_{HML}	-0.04 [-1.87]	-0.11 [-4.89]	-0.08 [-3.47]	-0.00 [-0.17]	-0.06 [-2.66]	-0.02 [-0.44]
β_{RMW}	0.05 [2.06]	-0.08 [-3.66]	0.05 [2.00]	0.11 [5.36]	0.12 [5.09]	0.07 [1.82]
β_{CMA}	-0.15 [-4.31]	-0.02 [-0.58]	0.03 [0.79]	0.30 [10.21]	0.25 [7.48]	0.40 [7.39]
β_{UMD}	-0.03 [-2.38]	-0.02 [-1.62]	0.02 [1.35]	-0.01 [-0.56]	0.01 [1.01]	0.04 [2.15]
Panel C: Average number of firms (n) and market capitalization (me)						
n	731	605	485	592	659	
me (\$10 ⁶)	1630	1381	1825	2022	2427	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EPI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.32 [3.96]	0.37 [4.66]	0.32 [4.10]	0.28 [3.45]	0.22 [2.83]	0.19 [2.43]
Quintile	NYSE	EW	0.55 [6.96]	0.65 [8.83]	0.54 [8.36]	0.44 [6.96]	0.36 [6.08]	0.29 [5.00]
Quintile	Name	VW	0.29 [3.61]	0.34 [4.14]	0.28 [3.50]	0.24 [2.93]	0.19 [2.39]	0.16 [2.05]
Quintile	Cap	VW	0.23 [3.00]	0.28 [3.59]	0.24 [3.12]	0.20 [2.50]	0.17 [2.20]	0.14 [1.80]
Decile	NYSE	VW	0.32 [3.32]	0.36 [3.80]	0.28 [2.97]	0.23 [2.46]	0.23 [2.49]	0.20 [2.15]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.28 [3.49]	0.34 [4.24]	0.29 [3.68]	0.26 [3.32]	0.21 [2.75]	0.19 [2.54]
Quintile	NYSE	EW	0.35 [4.10]	0.47 [5.85]	0.36 [5.14]	0.31 [4.49]	0.20 [3.02]	0.16 [2.60]
Quintile	Name	VW	0.25 [3.13]	0.30 [3.67]	0.24 [3.05]	0.22 [2.74]	0.17 [2.24]	0.16 [2.06]
Quintile	Cap	VW	0.20 [2.54]	0.24 [3.13]	0.20 [2.65]	0.18 [2.31]	0.15 [2.03]	0.14 [1.80]
Decile	NYSE	VW	0.27 [2.86]	0.31 [3.31]	0.23 [2.53]	0.21 [2.26]	0.20 [2.22]	0.19 [2.05]

Table 3: Conditional sort on size and EPI

This table presents results for conditional double sorts on size and EPI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EPI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EPI and short stocks with low EPI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EPI Quintiles					EPI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.39	0.72	0.95	0.96	0.97	0.60	0.69	0.59	0.52	0.38	0.33
		[1.39]	[2.73]	[3.61]	[3.89]	[4.11]	[5.44]	[6.47]	[6.06]	[5.30]	[4.17]	[3.64]
	(2)	0.51	0.77	0.87	0.96	0.92	0.41	0.51	0.38	0.32	0.27	0.23
		[2.04]	[3.22]	[3.65]	[4.17]	[4.21]	[3.97]	[5.14]	[4.21]	[3.54]	[3.00]	[2.55]
	(3)	0.62	0.58	0.80	0.83	0.94	0.32	0.39	0.29	0.25	0.19	0.17
	[2.86]	[2.72]	[3.58]	[4.09]	[4.79]	[3.54]	[4.46]	[3.52]	[3.05]	[2.30]	[2.04]	
(4)	0.50	0.61	0.85	0.75	0.79	0.28	0.35	0.23	0.21	0.04	0.04	
	[2.55]	[3.03]	[4.28]	[3.96]	[4.35]	[3.27]	[4.02]	[2.96]	[2.68]	[0.52]	[0.55]	
(5)	0.49	0.49	0.53	0.51	0.71	0.23	0.25	0.22	0.18	0.18	0.16	
	[2.94]	[2.78]	[3.10]	[3.14]	[4.53]	[2.39]	[2.64]	[2.32]	[1.89]	[1.90]	[1.61]	
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EPI Quintiles					EPI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	341	342	342	340	334	26	29	33	25	24	
	(2)	97	95	96	96	94	50	49	51	50	48	
	(3)	69	69	69	69	69	86	85	87	88	89	
	(4)	60	60	60	60	60	186	189	199	196	200	
(5)	56	56	56	56	55	1308	1408	1476	1529	1779		

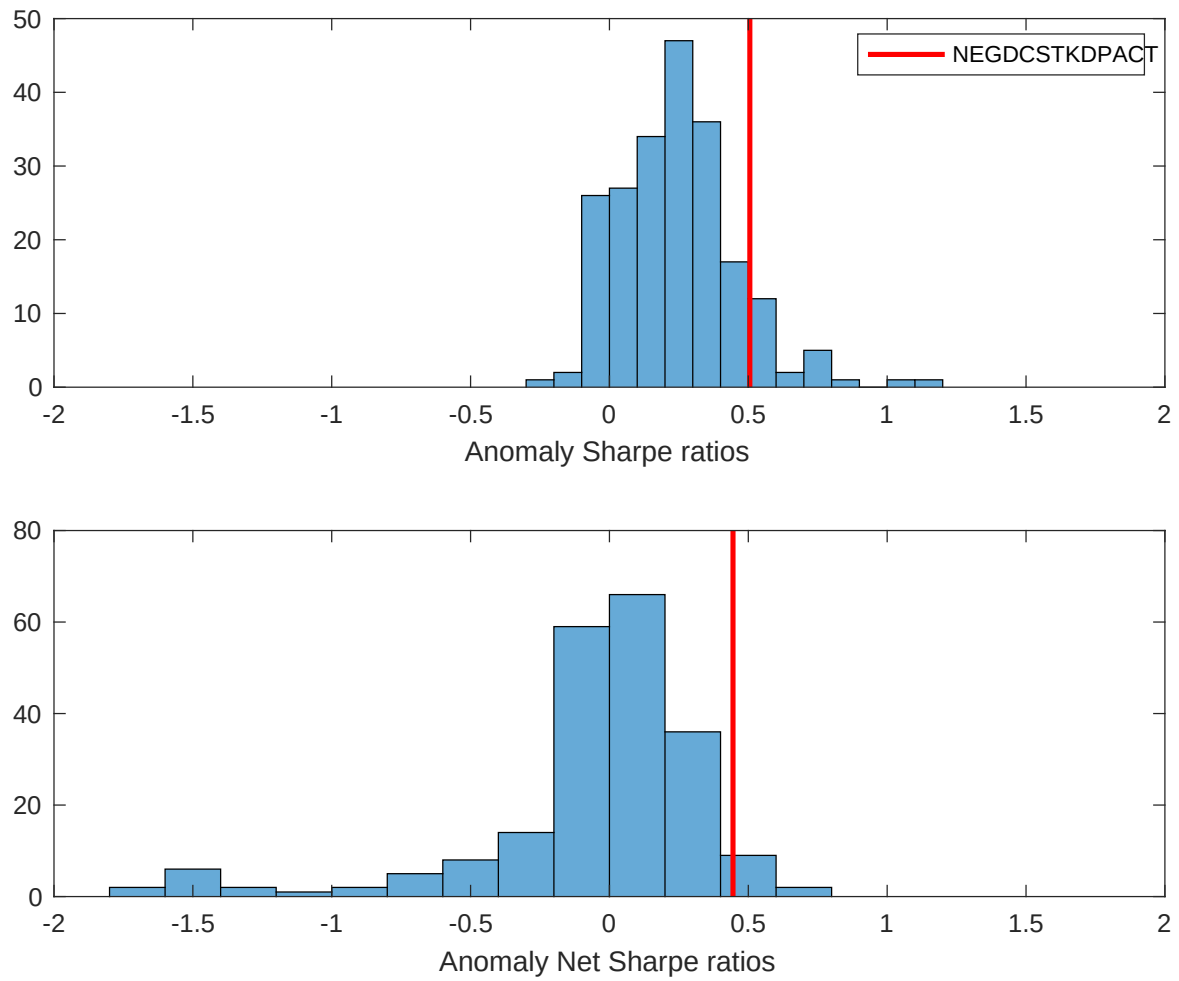


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EPI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

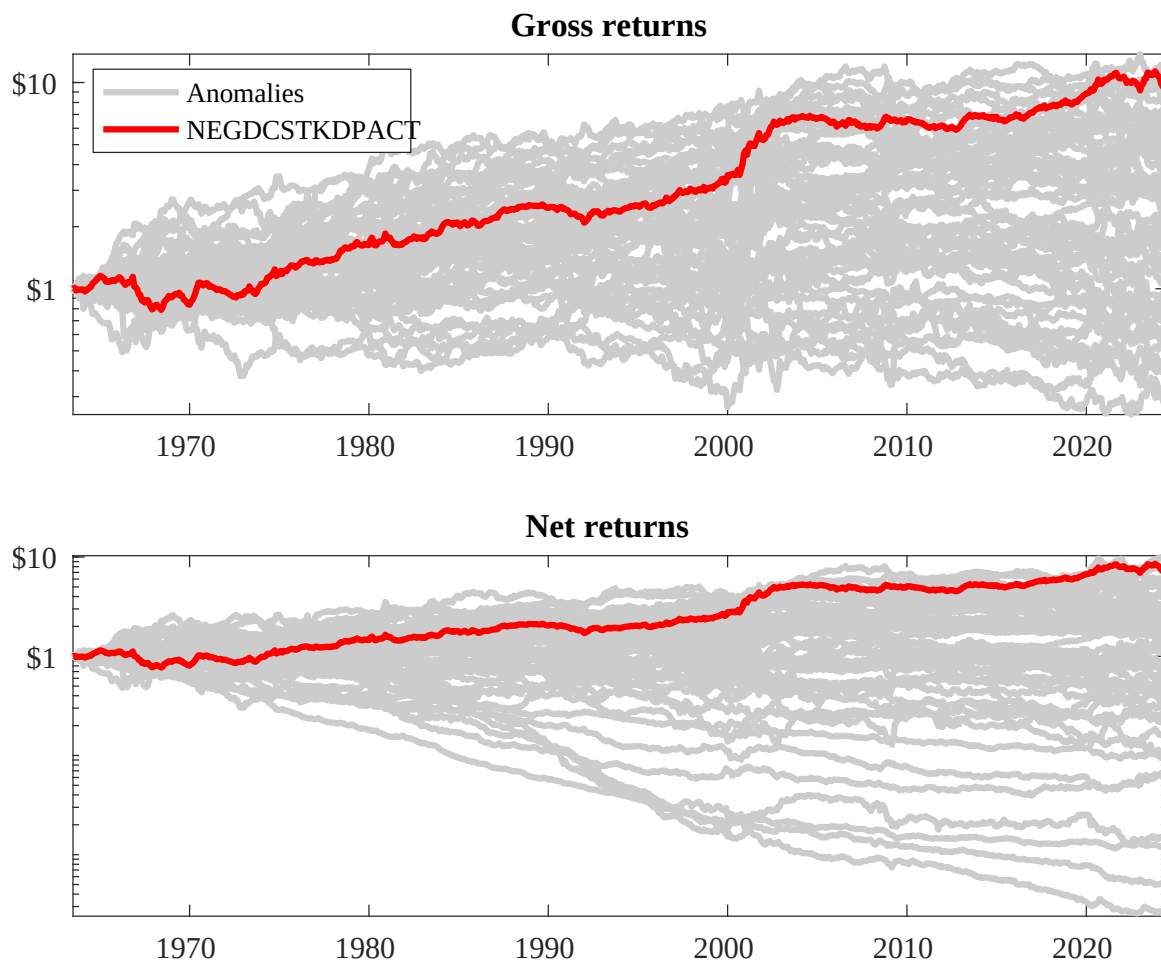
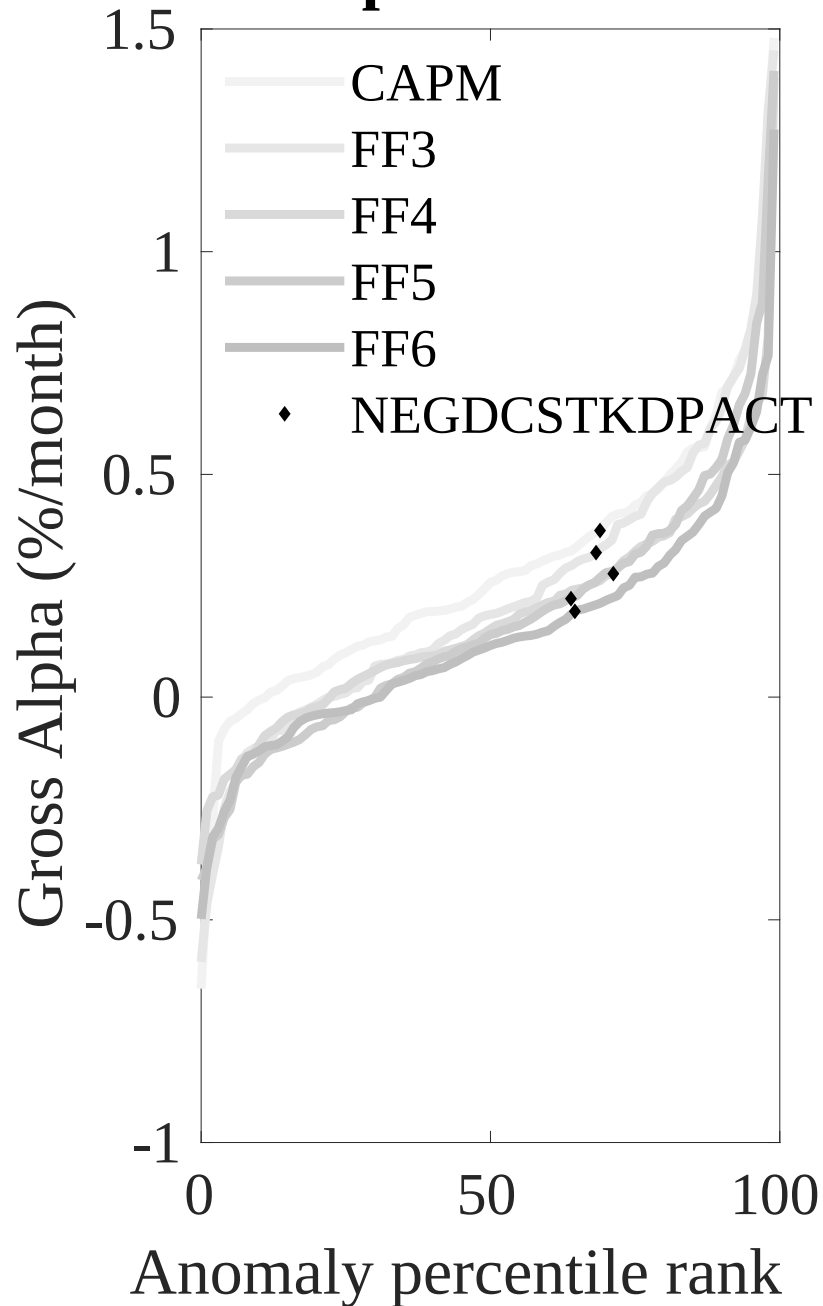


Figure 3: Dollar invested.

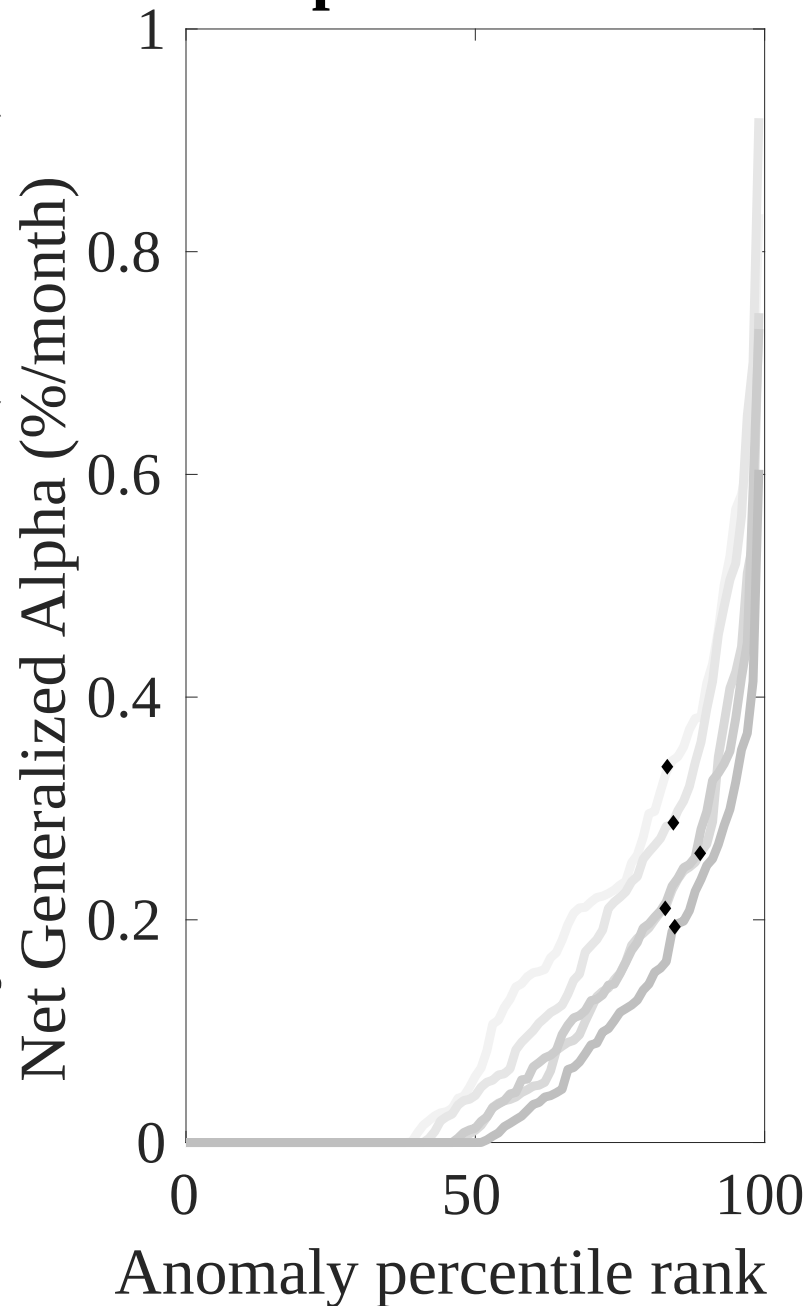
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EPI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



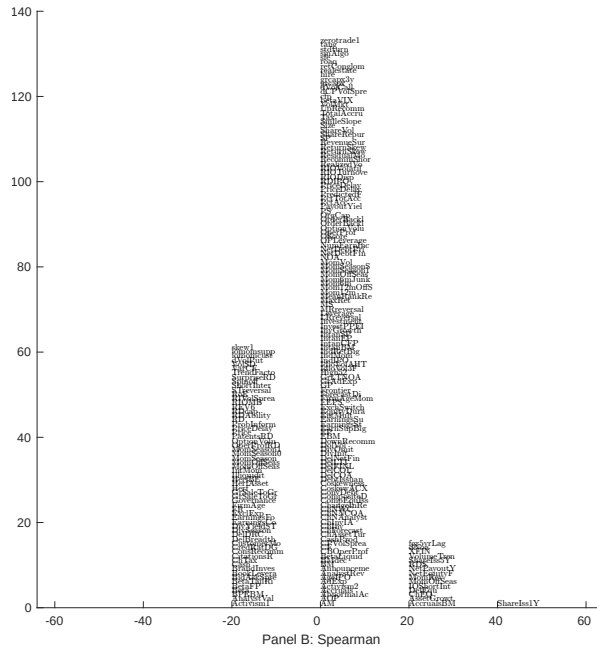
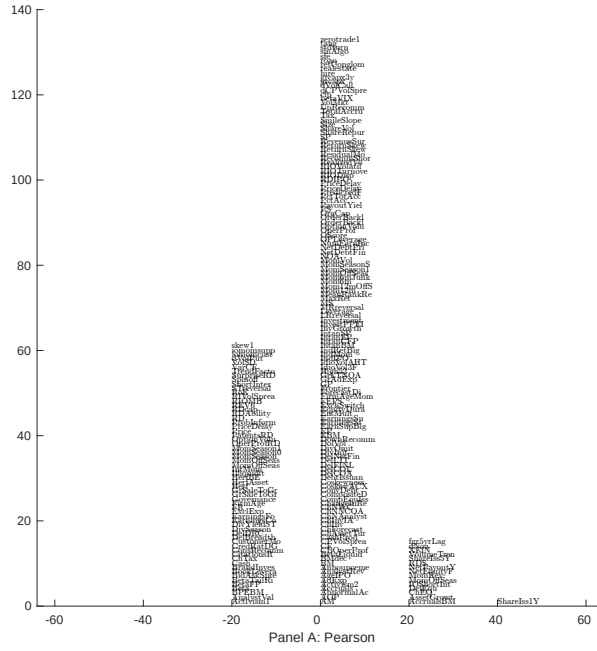


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with EPI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

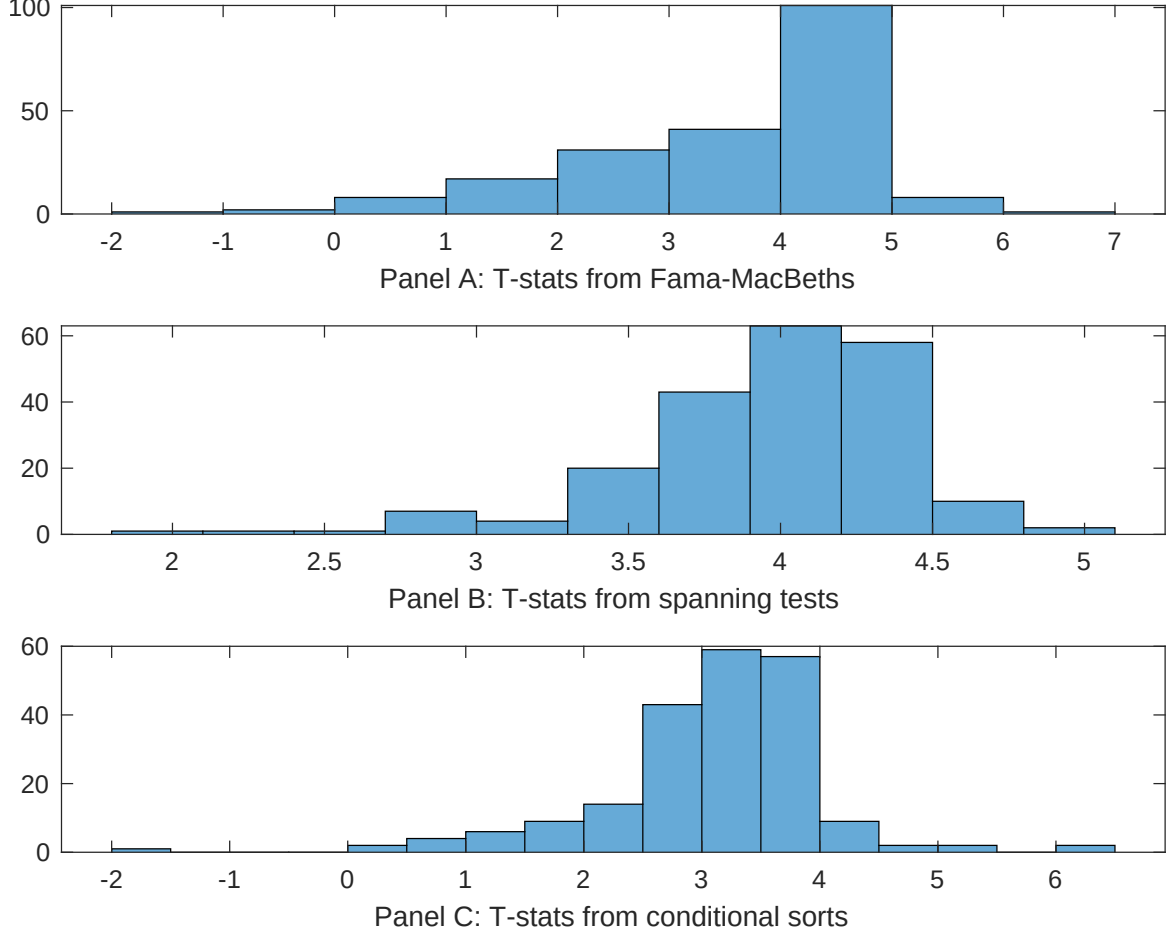


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EPI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EPI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EPI}EPI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EPI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EPI. Stocks are finally grouped into five EPI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EPI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EPI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EPI}EPI_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [5.98]	0.12 [5.43]	0.17 [6.64]	0.13 [6.37]	0.13 [5.61]	0.13 [5.42]	0.14 [5.00]
EPI	0.74 [3.87]	0.45 [2.07]	0.71 [3.32]	0.69 [3.66]	0.81 [3.92]	0.75 [3.47]	0.44 [1.90]
Anomaly 1	0.14 [6.01]						0.85 [3.78]
Anomaly 2		0.21 [1.88]					0.12 [2.60]
Anomaly 3			0.38 [3.21]				-0.25 [-0.16]
Anomaly 4				0.27 [4.73]			0.27 [5.88]
Anomaly 5					0.12 [3.17]		0.98 [1.63]
Anomaly 6						0.14 [2.45]	-0.15 [-1.85]
# months	715	715	720	715	720	630	623
$\bar{R}^2(\%)$	0	1	0	1	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EPI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EPI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.18 [2.33]	0.23 [2.97]	0.18 [2.35]	0.12 [1.63]	0.19 [2.48]	0.23 [2.84]	0.19 [2.43]
Anomaly 1	30.92 [8.10]						13.48 [2.64]
Anomaly 2		19.34 [6.52]					12.96 [3.32]
Anomaly 3			33.34 [8.18]				38.69 [6.22]
Anomaly 4				27.98 [6.86]			14.80 [2.93]
Anomaly 5					13.83 [3.57]		-18.62 [-3.26]
Anomaly 6						7.35 [1.96]	-17.37 [-3.79]
mkt	-0.07 [-0.04]	2.16 [1.15]	0.29 [0.16]	0.43 [0.24]	-1.06 [-0.57]	-0.41 [-0.20]	0.05 [0.02]
smb	-0.17 [-0.07]	1.75 [0.65]	-2.46 [-0.95]	0.45 [0.17]	-1.52 [-0.57]	5.96 [1.84]	-1.23 [-0.40]
hml	-1.24 [-0.36]	-7.95 [-2.13]	-4.43 [-1.26]	1.87 [0.53]	-2.15 [-0.59]	0.03 [0.01]	-3.81 [-1.03]
rmw	-9.97 [-2.45]	-4.49 [-1.13]	8.35 [2.38]	-3.24 [-0.84]	8.04 [2.21]	9.27 [2.10]	2.28 [0.49]
cma	19.82 [3.54]	22.74 [3.98]	6.73 [1.04]	25.41 [4.61]	25.12 [3.83]	22.65 [3.76]	-6.08 [-0.88]
umd	2.81 [1.57]	5.76 [3.17]	3.65 [2.05]	2.40 [1.32]	4.59 [2.49]	5.15 [2.70]	3.79 [2.09]
# months	728	728	732	728	732	630	626
$\bar{R}^2(\%)$	22	20	22	20	16	12	24

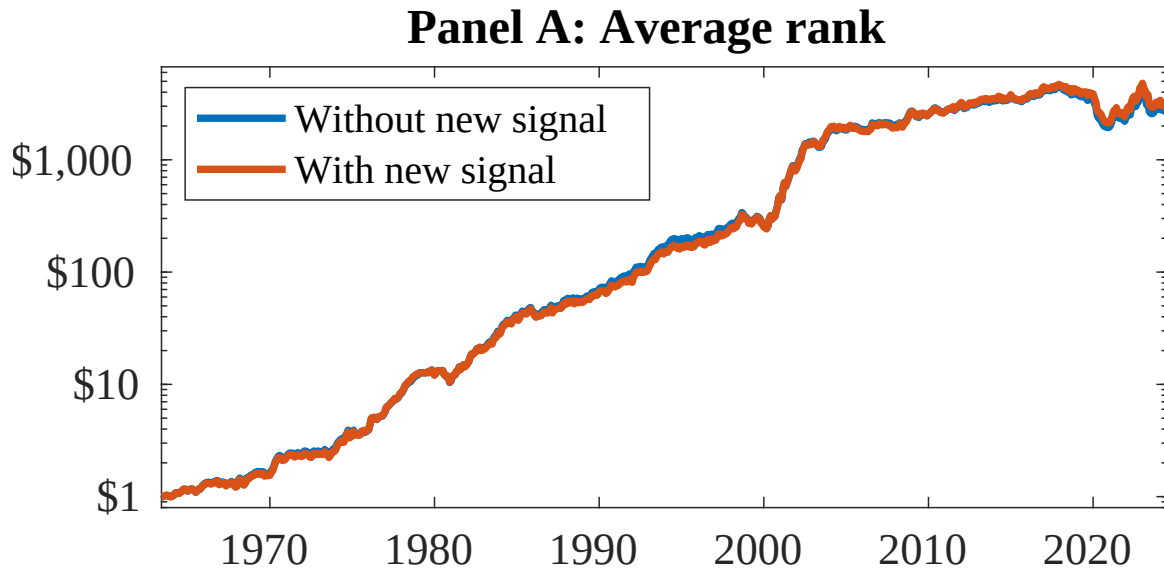


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EPI. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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